

Thomas Ngo

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Portfolio | LinkedIn | GitHub

EDUCATION

University of Alberta

Bachelor of Science: Computing Science (Honors)

Edmonton, AB

Sep 2023 – Expected June 2027

- **GPA: 3.86 / 4.00**
- **Honors & Awards:** International Admission Scholarship, Regional Excellence Scholarship.
- **Relevant Coursework:** Machine Learning I & II, Algorithms, Visual Recognition (Computer Vision), Web Applications & Architecture, Search & Planning in AI, Mathematical Statistics, Numerical Methods.

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript, TypeScript, C/C++, C#, SQL, R, LaTeX, HTML/CSS.
- **AI & Data Science:** Deep Learning, Natural Language Processing, Computer Vision (OpenCV), LLM Integration, Scikit-Learn, PyTorch, NumPy, Pandas, Time Series Analysis.
- **Frameworks & Web:** React, Node.js, Express, Spring Boot, Tailwind CSS, JUnit, RabbitMQ.
- **Cloud & DevOps:** Docker, Git, CI/CD Pipelines, Firebase, Linux/Unix.
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, SQLite3, Firestore.

PROFESSIONAL EXPERIENCE

Viettel Military Industry and Telecoms Group

Software Engineering Intern (Web Programming)

Remote / Hanoi, Vietnam

May 2026 – July 2026

- Completed intensive training in enterprise-scale system architecture and full-stack development, then applied it by independently designing and delivering a production-style microservices platform as the core internship deliverable (detailed under Projects).
- Made and defended key architectural decisions — service boundaries, data consistency, concurrency control — in technical design reviews with senior engineers.
- Applied **Agile/Scrum** methodologies to the software development lifecycle, collaborating with senior mentors to pitch technical innovations to executive leadership.

Cosmos Collision Ltd

Software & Digital Operations Intern

Calgary, Canada

May 2026 – August 2026

- Authored standard operating procedures (SOP) and managed daily operations for 3D scanning, 3D printing, and CNC carving equipment to streamline internal fabrication workflows.
- Built a Chrome extension that auto-extracts data from collision estimate PDFs and fills out Mitchell intake forms, cutting manual data entry across contact/vehicle/insurance sections and reducing transcription errors.

SELECTED PROJECTS

Agri Rescue Platform

Java, Spring Boot, React, TypeScript, PostgreSQL, RabbitMQ, Docker

May 2026 – August 2026

- Designed a modular **REST API** across **7 microservices** with **JWT-based authentication**, using an **event-driven** architecture (**RabbitMQ**) and the outbox pattern for reliable inter-service communication.
- Engineered concurrency-safe transactions with **pessimistic/optimistic** locking and a **finite state machine** enforcing a valid produce lifecycle, behind a **JWT-secured Spring Cloud Gateway**.
- Built a **React/TypeScript** dashboard with **LLM-powered** (Google's Gemini API) analytics assistant, and containerized the system with **Docker Compose** over a **schema-per-service PostgreSQL** architecture.

Android Event Lottery Application | Java, Firebase, CI/CD

Oct 2025 – Dec 2025

- Co-developed a native Android application using **Java** and **Android SDK**, building modular UI layouts with **XML** and **Fragments** for scalable navigation across 500+ prospective users.
- Implemented real-time data synchronization using **Firebase Firestore** (NoSQL) for live event tracking, participant lottery pools, and instant **QR-code** check-in validation.
- Engineered Role-Based Access Control (**RBAC**) workflows for admin, organizer, and participant roles, integrating dynamic push notifications and automated lottery selection algorithms.

- Architected automated **CI/CD pipelines** with **JUnit** unit testing, ensuring zero-regression code updates and reducing deployment overhead by **30%**.

Optimizer Animated Visualizer | *React, TypeScript, Python, Node.js, Railway*

May 2026

- Developed a full-stack benchmarking platform to simulate and visualize 12 ML optimizer-kernel combinations; engineered a **Python/NumPy** engine to execute high-performance numerical optimization experiments.
- Built an interactive **React/TypeScript** dashboard featuring real-time playback controls and dynamic animations using **Recharts**, enabling granular analysis of learning curve convergence.
- Designed a modular **RESTful API** to bridge the numerical backend with the frontend, deploying the system via **Railway** with an automated CI/CD pipeline for rapid iteration.

EXTRACURRICULAR & LEADERSHIP

UAlberta Book Club | *VP of Marketing*

May 2025 – April 2026

- Spearheaded multi-platform marketing strategies and social media campaigns across Instagram and campus channels to boost brand visibility.
- Designed graphics, promotional assets, and event announcements, driving consistent member participation and expanding active club engagement.
- Collaborated with executive team members to plan and promote large-scale events, book drives, and community meetups.

UofA Computing Science Department | *Student Volunteer*

Oct 2024

- Represented the Department of Computing Science at prospective student events, engaging with hundreds of visitors to provide tailored guidance on degree paths.
- Addressed inquiries regarding undergraduate course structures, faculty resources, research opportunities, and student life to assist prospective students in their academic decision-making.